

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, C++, Java, JavaScript, TypeScript, SQL, Bash

ML & Autodiff

JAX, PyTorch, NumPy, reverse-mode autodiff (from scratch), normalizing flows, diffusion, scikit-learn, HuggingFace Transformers

Optimization

GD / SGD convergence theory, natural-gradient (NGVI), variational inference, Fisher information, learning-rate tuning

Performance

Performance optimization, Linux profiling, OpenMP / SIMD concepts, cache-aware design, Apache Spark / PySpark, distributed-data concepts

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability across releases and remediated **20+ VAPT findings**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by PostgreSQL, and integrated LLM-powered context-aware replies into a chat product, iterating on prompt design and response-quality heuristics for a real customer-facing experience.

Projects

Autodiff Engine & Generative Models (from scratch)

github.com/Deva-1903/UMASS_CICS_689

JAX, NumPy, PyTorch | UMass CS 689 Advanced ML

- Implemented a **matrix-based reverse-mode automatic differentiation engine from scratch** in NumPy, computation graphs, topological-sort backpropagation, and operators including matmul, linear solve, and log-determinant, **validated against JAX** to machine precision.
- Implemented a **normalizing flow with coupling layers** and a **DDPM diffusion model** from scratch in **JAX** (custom forward/reverse processes, ELBO via variational inference); benchmarked 5 architectures across 3 optimizers on CIFAR-10 and derived GD/SGD convergence guarantees.

NGVI Curvature-Variance Study (Optimization)

github.com/Deva-1903/ngvi-curvature-variance

PyTorch (CUDA 12.6), NumPy | UMass CS 651 Optimization (team)

- Built a PyTorch testbed comparing Adam, closed-form **natural-gradient (NGVI)**, and EMA Diagonal-Fisher optimizers; instrumented ELBO, Hessian condition number, and gradient variance to separate curvature from Monte-Carlo-noise bottlenecks (antithetic sampling cut gradient variance **10x**).
- Showed NGVI's ELBO gain over Adam grows with posterior ill-conditioning (κ from 180 to 2566 maps to +0.28 to +2.32 nats), and that closed-form NGVI beats an EMA Diagonal-Fisher method in wall-clock despite $O(D^3)$ Fisher-inversion cost.

Spark ETL Performance Optimization

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, SQL, Python | 1.4B-row dataset

- Cut data-processing runtime **25%** (17 to 19 min down to 14.3 min) on a **1.4B-row** dataset via broadcast joins, adaptive query execution, and partition tuning, and reduced analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x**, profiling stage timings via Spark UI.

AgenticSearch - Modular ML Pipeline & Developer Tooling

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite

- Built a modular, multi-stage ML pipeline (typed planning, extraction, verification) hardened with **150+ automated tests** and pipeline instrumentation, the kind of developer tooling that streamlines applied ML work.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.